National Institute of Business Management School of Computing and Engineering

Higher National Diploma in Software Engineering

HNDSE25.2F Internet of Things(IoT)

Proposal report

AI-Powered Agribot for Smart Detection of Nutrient Deficiencies in Tomato and Potato Crops Using TinyML and IoT

Submitted by

D.G.C Minidu	KAHNDSE25.2F -005
J.P.K.S Lakmal	KAHNDSE25.2F -014
K.P.G.M.M Rajapaksha	KAHNDSE25.2F -031
G.W.D.B Weerasekara	KAHNDSE25.2F -032

Submitted date.

04/09/2026

Contents

Executive Summary

Brief Overview of the Project

Agriculture is increasingly being transformed through the integration of intelligent technologies such as Artificial Intelligence (AI) and the Internet of Things (IoT). This project proposes the design and development of an AI-powered Agribot capable of detecting nutrient deficiencies in tomato and potato crops using image-based classification and embedded machine learning techniques.

The system utilizes the ESP32-CAM as the central processing unit, enabling real-time image capture and on-device inference. A Convolutional Neural Network (CNN) is trained using Edge Impulse and deployed on the device for classification tasks.

The Agribot is equipped with a multi-tank spraying mechanism and a rotating nozzle system that enables targeted delivery of nutrients based on detected deficiencies. Additionally, environmental sensors are used to monitor plant growth conditions. Communication is achieved using MQTT, enabling real-time data transmission and monitoring.

Objectives and Expected Outcomes

Objectives:

- Develop an autonomous robotic platform for agricultural monitoring
- Implement AI-based detection of nutrient deficiencies
- Enable automated, targeted spraying using a multi-nozzle system
- Integrate environmental sensing for plant growth monitoring
- Facilitate real-time communication using MQTT

Expected Outcomes:

- Accurate classification of plant deficiencies (>80%)
- Reduced manual labor in agriculture
- Improved precision in fertilizer application
- Enhanced crop productivity and sustainability

Project Background

Introduction

Agriculture is a cornerstone of economic stability and food security in developing regions. However, inefficiencies persist due to reliance on manual inspection and generalized fertilizer application. Nutrient deficiencies—particularly nitrogen (N), phosphorus (P), and potassium (K)—significantly impact crop health, yield, and quality. Early detection is crucial but often inaccurate due to human subjectivity.

Recent advancements in embedded AI and IoT enable **on-device intelligence**, reducing latency and dependency on cloud infrastructure. This project proposes an autonomous Agribot that leverages TinyML for real-time leaf analysis and performs precise interventions.

Problem Domain

One of the key challenges in agriculture is the identification of nutrient deficiencies in crops. Essential nutrients such as nitrogen (N), phosphorus (P), and potassium (K) play critical roles in plant growth. Deficiencies in these nutrients can lead to:

- Reduced photosynthesis
- Poor plant development
- Decreased yield

Manual identification is often unreliable due to:

- Human error
- Lack of expertise
- Delayed detection

Motivation

The motivation for this project arises from the need to:

- Improve early detection of plant stress
- Automate corrective actions

- Reduce dependency on manual labor
- Introduce low-cost smart farming solutions

Project Aim and Objectives

Aim and Objectives

Aim

To design and implement an AI-enabled Agribot that detects nutrient deficiencies in tomato and potato leaves and performs automated, targeted spraying using a multi-nozzle system.

Objectives

- Develop a mobile robotic platform using an embedded controller
- Capture and preprocess leaf images in real time
- Train a CNN-based classification model using TinyML
- Classify conditions into: Healthy, Nitrogen, Phosphorus, Potassium, Water Stress
- Design a 4-tank spraying system with controlled actuation
- Integrate environmental sensors for growth monitoring
- Implement MQTT-based communication for telemetry and control
- Evaluate system performance (accuracy, latency, power consumption)

Scope and Limitations

Scope

- Crops: Tomato and Potato
- Conditions: 5 classes per crop (10 total)
- On-device inference using embedded hardware
- Semi-controlled field testing

Limitations

- Visual mapping of diseases to deficiency-like categories (approximation)
- Dependence on lighting conditions for image capture
- Limited model size due to embedded constraints

Methodology / Approach

Steps or Phases of the Project

Phase 1: Research & Planning

- Literature review and system design

Phase 2: Dataset Preparation

- Collect images from datasets and online sources
- Label images based on visual symptoms

Phase 3: AI Model Development

- Train CNN model using Edge Impulse
- Optimize for embedded deployment

Phase 4: Hardware Development

- Assemble robot chassis
- Integrate motors, sensors, and pumps

Phase 5: System Integration

- Combine AI model with hardware
- Implement control logic

Phase 6: Testing & Evaluation

- Evaluate accuracy and performance

Literature Review

AI in Agriculture

Recent studies demonstrate the effectiveness of deep learning models in plant disease detection. CNN-based approaches have achieved accuracy levels exceeding 90% in controlled datasets.

TinyML Applications

TinyML enables machine learning inference on low-power devices such as microcontrollers. This reduces reliance on cloud computing and enables real-time decision-making.

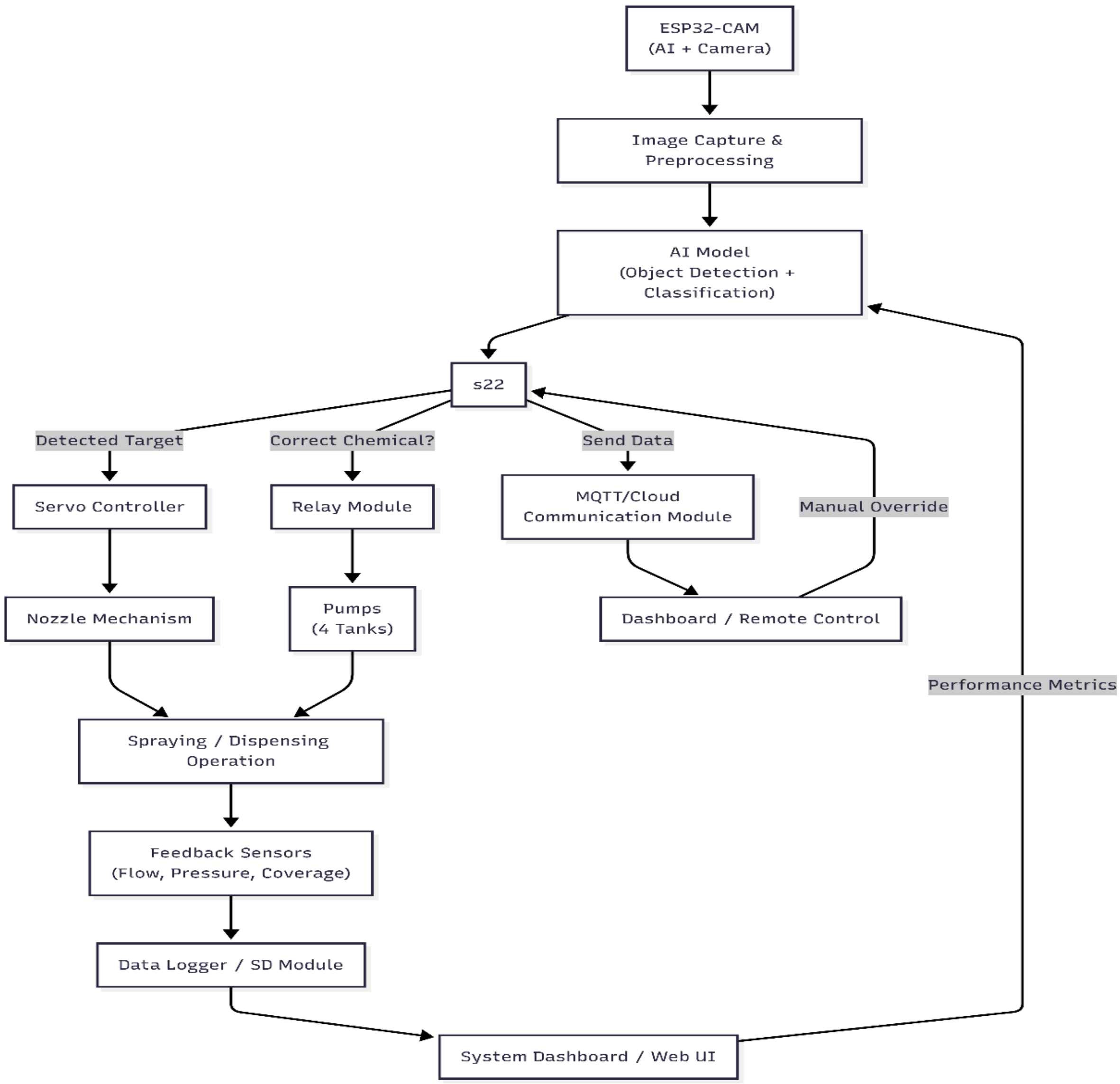
IoT in Smart Farming

IoT systems facilitate:

- Real-time monitoring
- Remote control
- Data-driven insights

Tools, Technologies, or Software

System Architecture



Component Interaction

Component	Function
ESP32-CAM	Processing + image capture
CNN Model	Classification

Component	Function
Sensors	Environmental monitoring
Pumps	Liquid delivery
MQTT	Communication

Methodology

Dataset Preparation

- Source: Public datasets + curated web images
- Strategy: Visual re-labeling based on symptoms
- Classes: 10 (Tomato/Potato × 5 conditions)

Table 1: Class Definitions

Class	Visual Indicator
Healthy	Uniform green
Nitrogen	Yellowing
Phosphorus	Purple tint
Potassium	Brown edges
Water Stress	Wilting/drying

Flowchart

AI-Pipeline Flow Chart

Images → Preprocessing → CNN Training → Model Export → Deployment

AI-Model Design

CNN Architecture

Input (96x96)
↓
Conv Layer
↓
MaxPooling
↓
Conv Layer
↓
Flatten
↓
Dense Layer
↓
Softmax Output

Data Preprocessing

- Resize: 96×96
- Normalize pixel values
- Remove noise and duplicates
- Augmentation: rotation, flipping, brightness variation

Model Development

Platform: Edge Impulse

Model Architecture:

- Conv2D → ReLU → MaxPool
- Conv2D → ReLU → MaxPool
- Flatten → Dense → Softmax

Training Configuration

Table 2: Training Parameters

Parameter	Value
Epochs	30
Batch Size	16
Split	70/20/10
Input Size	96×96

Deployment

- Export as Arduino library
- Upload to ESP32-CAM
- Real-time inference from camera feed

Hardware Design

Controller

- ESP32-CAM

Mobility

- 4WD chassis + DC motors
- Motor driver (L298N)

Spraying System

- 4 tanks (N, P, K, Water)
- 4 pumps
- 4-channel relay

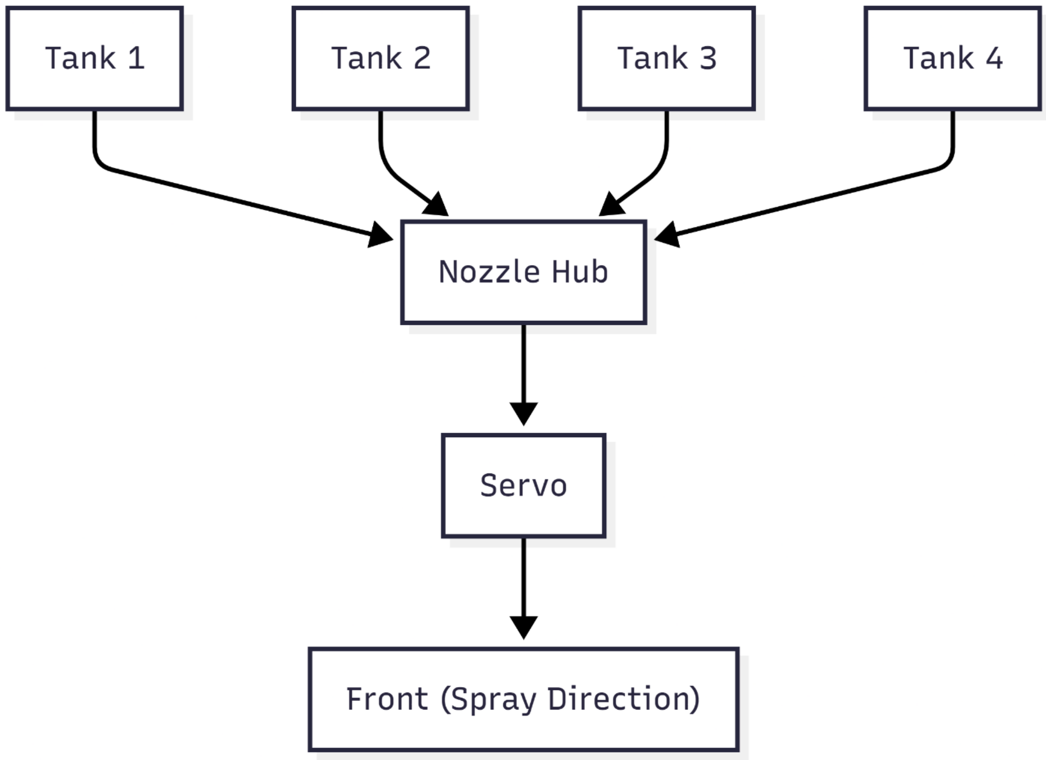


Figure 2: Mechanical Layout

Nozzle Mechanism Design

- 4 nozzles mounted on a rotating disc
- Servo enables circular motion
- Secondary servo enables vertical tilt

IoT Communication

Protocol: MQTT

Topics:

- agribot/detection
- agribot/action
- agribot/status

Growth Monitoring

Sensors:

- Soil Moisture
- Temperature/Humidity

Rule Logic:

IF moisture < threshold → Water Stress
IF temp high → Heat Stress
IF healthy + optimal → Good Growth

System Workflow

Capture → Classify → Decide → Rotate → Spray → Report (MQTT)

Testing and Evaluation

Metrics:

- Accuracy
- Latency
- Power consumption

Test Cases:

- Controlled leaf samples
- Different lighting conditions

Pseudocode

```
START
Capture Image
Run AI Model
IF Deficiency == Nitrogen
  Activate Tank 1
ELSE IF Deficiency == Phosphorus
  Activate Tank 2
ELSE IF Deficiency == Potassium
  Activate Tank 3
ELSE IF Deficiency == Water Stress
  Activate Tank 4
ELSE
  No Action
Rotate Nozzle
Spray
Send Data via MQTT
END
```

Project Deliverables

Expected Outputs

Deliverable	Description
Prototype	Fully functional Agribot
AI Model	Trained CNN model
Software	Embedded system code
Report	Final project documentation
Presentation	Slides and demo

Timeline

Gantt Chart Placeholder

Week	Activity
1–2	Research
3–4	Data
5–6	AI Model
7–8	Hardware
9	Integration
10	Testing
11–12	Report

Expected Results

- Accurate classification (>80%)
- Automated spraying
- Reduced labor
- Improved efficiency

Limitations

- Dataset limitations
- Lighting dependency
- Simplified deficiency mapping

Future Work

- Mobile app integration
- Cloud analytics

References

1. Edge Impulse, 2024. *Edge Impulse documentation*. Available at: https://docs.edgeimpulse.com (Accessed: 10 April 2026).

2. Kaggle, 2024. *Plant Village dataset*. Available at: https://www.kaggle.com (Accessed: 10 April 2026).

3. FAO, 2022. *Global agriculture outlook*. Available at: https://www.fao.org (Accessed: 10 April 2026).

4. Goodfellow, I., Bengio, Y. and Courville, A., 2016. *Deep learning*. Cambridge: MIT Press.

5. Russell, S. and Norvig, P., 2021. *Artificial intelligence: A modern approach*. 4th ed. Pearson.

6. Bishop, C.M., 2006. *Pattern recognition and machine learning*. Spring

Appendices

E1. Dataset Composition Summary

The dataset used for training the TinyML model consists of images collected from multiple open-source platforms and manually curated sources. The dataset was structured to ensure class balance and improve model generalization.

Class Label	Description	Number of Images
Tomato_Healthy	Healthy tomato leaves	500
Tomato_Nitrogen	Yellowing of older leaves	400
Tomato_Phosphorus	Purple/dark discoloration	400
Tomato_Potassium	Leaf edge burn	400
Potato_Healthy	Healthy potato leaves	500
Potato_Nitrogen	Pale/yellow leaves	400
Potato_Phosphorus	Dark/purple leaves	400
Potato_Potassium	Brown edges/scorching	400

Total Dataset Size: ~3400 images
To improve performance, the dataset was augmented using rotation, flipping, zooming, and brightness adjustments.

E2. Data Augmentation Parameters

To enhance the robustness of the AI model, data augmentation techniques were applied during training.

Parameter	Value
Rotation Range	±20°
Horizontal Flip	Enabled

Parameter	Value
Vertical Flip	Enabled
Zoom Range	0.8 – 1.2
Brightness Adjustment	±15%
Image Resolution	96 × 96 pixels

These transformations help the model perform well under real-world conditions such as varying lighting, camera angles, and environmental disturbances.

E3. Sensor Data Range and Threshold Values

The Agribot integrates environmental sensors to support AI-based decision-making. The following thresholds are used to assess plant health:

Parameter	Optimal Range	Deficiency Indicator
Soil Moisture	40% – 70%	< 30% (Water deficiency)
Temperature	20°C – 30°C	> 35°C (Stress condition)
Humidity	50% – 80%	< 40% (Dry environment)

Sensor data is continuously monitored and transmitted to the microcontroller for processing.

E4. AI Model Configuration

The AI model is developed using TinyML principles to ensure efficient edge deployment.

Parameter	Value
Model Type	Convolutional Neural Network (CNN)
Input Size	96 × 96 × 3

Parameter	Value
Epochs	50
Batch Size	16
Learning Rate	0.001
Framework	Edge Impulse
Deployment	ESP32-CAM

The model is optimized for low power consumption and real-time inference.

E5. Expected Model Performance Metrics

The expected performance of the trained model is evaluated using standard metrics:

Metric	Expected Value
Accuracy	85% – 92%
Precision	83%
Recall	80%
F1-Score	81%

These values are based on similar studies conducted in plant disease detection using edge AI systems.

E6. Decision Logic Mapping

The system integrates AI predictions with sensor data to make decisions:

AI Output	Sensor Condition	Action
Nitrogen Deficiency	Normal moisture	Spray Nitrogen solution
Phosphorus Deficiency	Normal temperature	Spray Phosphorus solution
Potassium Deficiency	Dry soil	Spray Potassium + Water
Healthy Plant	Optimal conditions	No action

E7. Liquid Distribution System Data

The robot uses a multi-tank spraying mechanism with controlled nozzles.

Tank	Content	Flow Rate
Tank 1	Nitrogen Solution	10 ml/sec
Tank 2	Phosphorus Solution	10 ml/sec
Tank 3	Potassium Solution	10 ml/sec
Tank 4	Water	15 ml/sec

Nozzles rotate in a circular motion to ensure uniform distribution.

E8. Power Consumption Estimation

Component	Voltage	Current
ESP32-CAM	5V	160 mA
Servo Motor	5V	500 mA
Water Pump	6V	300 mA

Component	Voltage	Current
Sensors	3.3V	50 mA

Total Estimated Consumption: ~1A
A rechargeable lithium-ion battery is recommended for portability.

E9. Growth Monitoring Indicators

The system evaluates plant growth based on visual and environmental parameters:

- Leaf color intensity
- Leaf size and spread
- Number of leaves
- Moisture consistency
- Growth rate over time

These indicators help determine whether the plant is developing normally or experiencing stress.